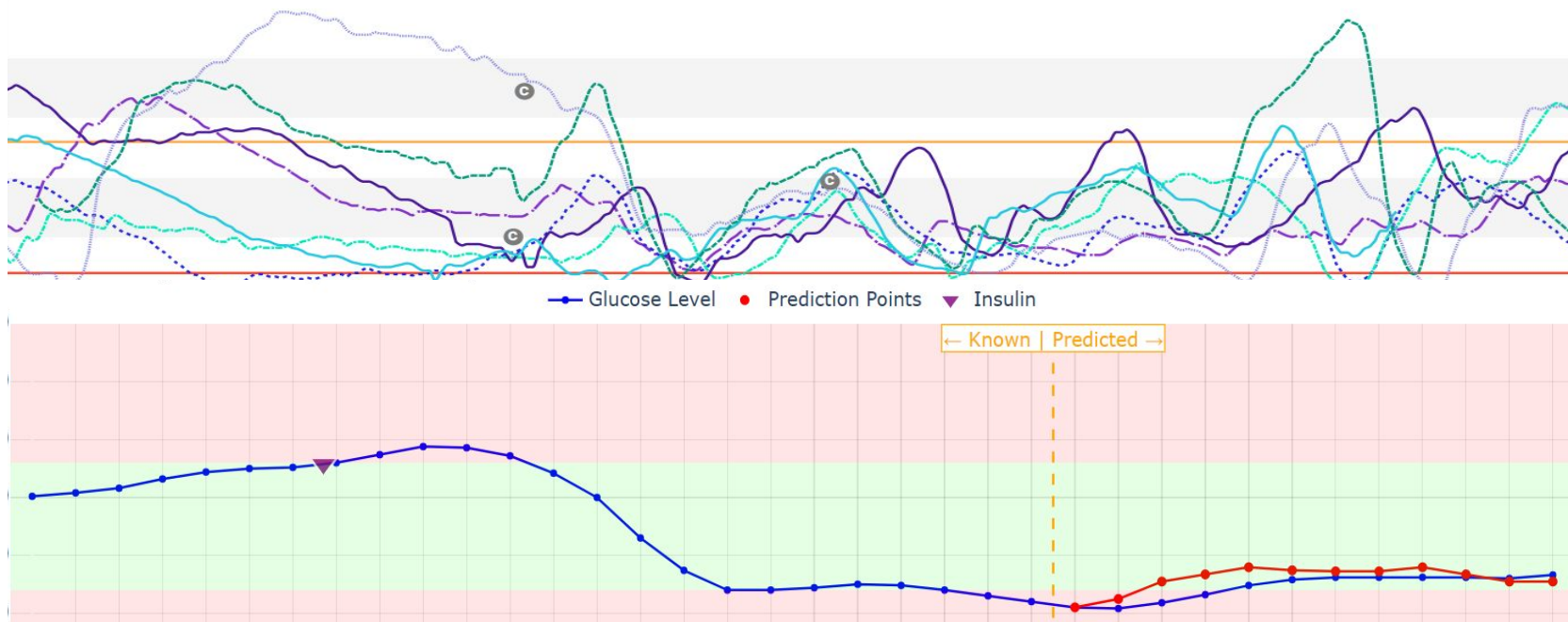


MULTIMODAL TRANSFORMER-BASED PREDICTION OF BIOMEDICAL TIME SERIES FROM WEARABLE SENSORS: A CASE STUDY IN NEXT-HOUR GLUCOSE FORECASTING

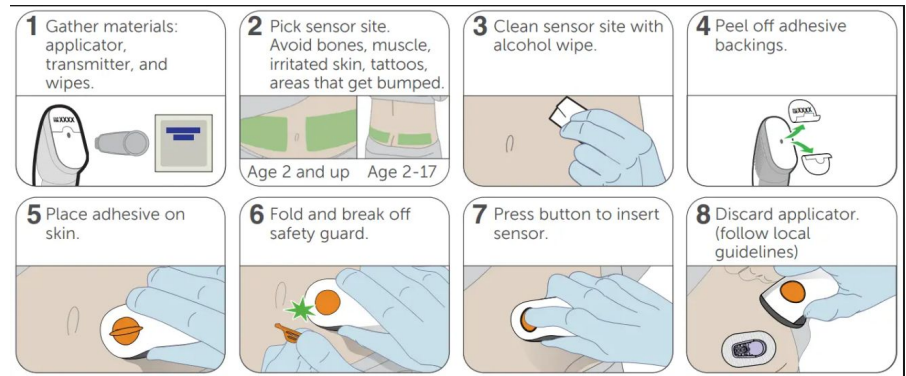


CGMs... what ARE they?

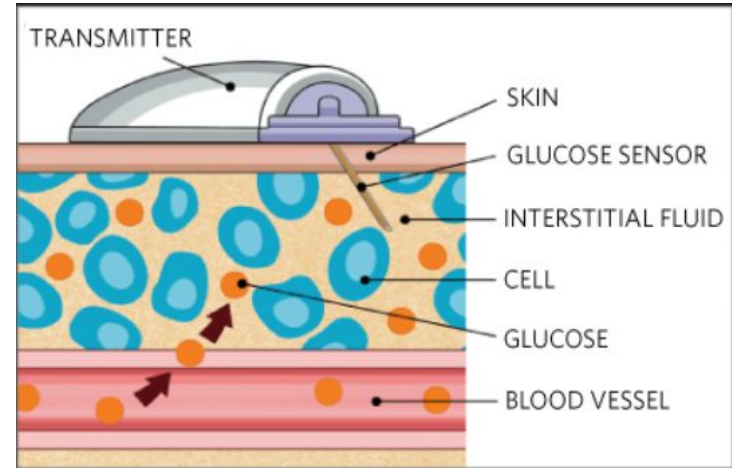
CGM- Continuous Glucose Monitoring systems are medical devices which are externally mounted on humans and measure the value of glucose every 5 min



This is also a sensor +transmitter



source: Dexcom G6 PRO Professional Continuous Glucose Monitoring Instruction Manual



source: <https://houstonendocrinecenter.com/what-are-continuous-glucose-monitoring-systems/>

CGMs... what do they give?

90 days | Fri Mar 4, 2022 - Wed Jun 1, 2022

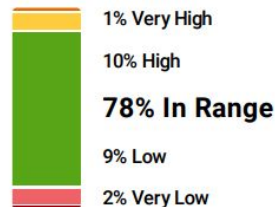
Patient

Glucose

Average Glucose

123 mg/dL

Time in Range



Sensor Usage

Days with CGM data

100%

90/90

Standard Deviation

48 mg/dL

GMI

6.3%

Target Range:

70-180 mg/dL

Avg. calibrations per day

0.4

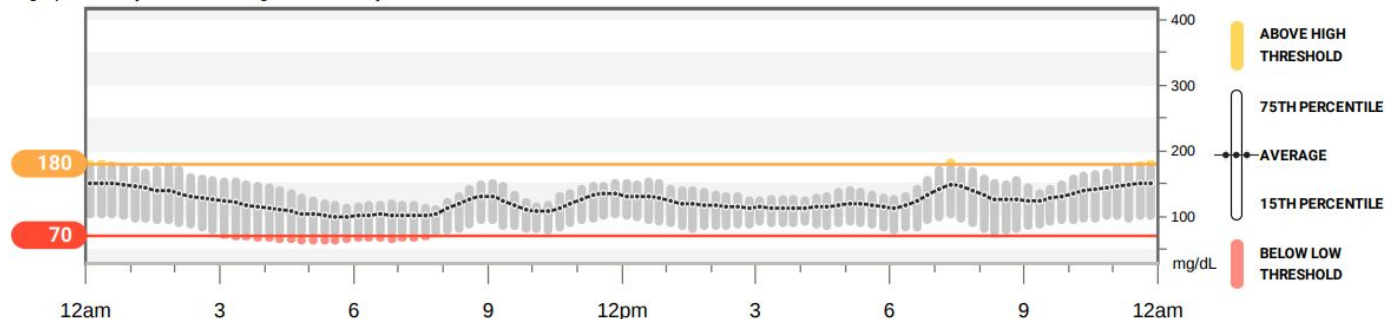
Top Patterns



Patient's best glucose day was March 23, 2022

Patient's glucose data was in the target range about 97% of the day.

This graph shows your data averaged over 90 days



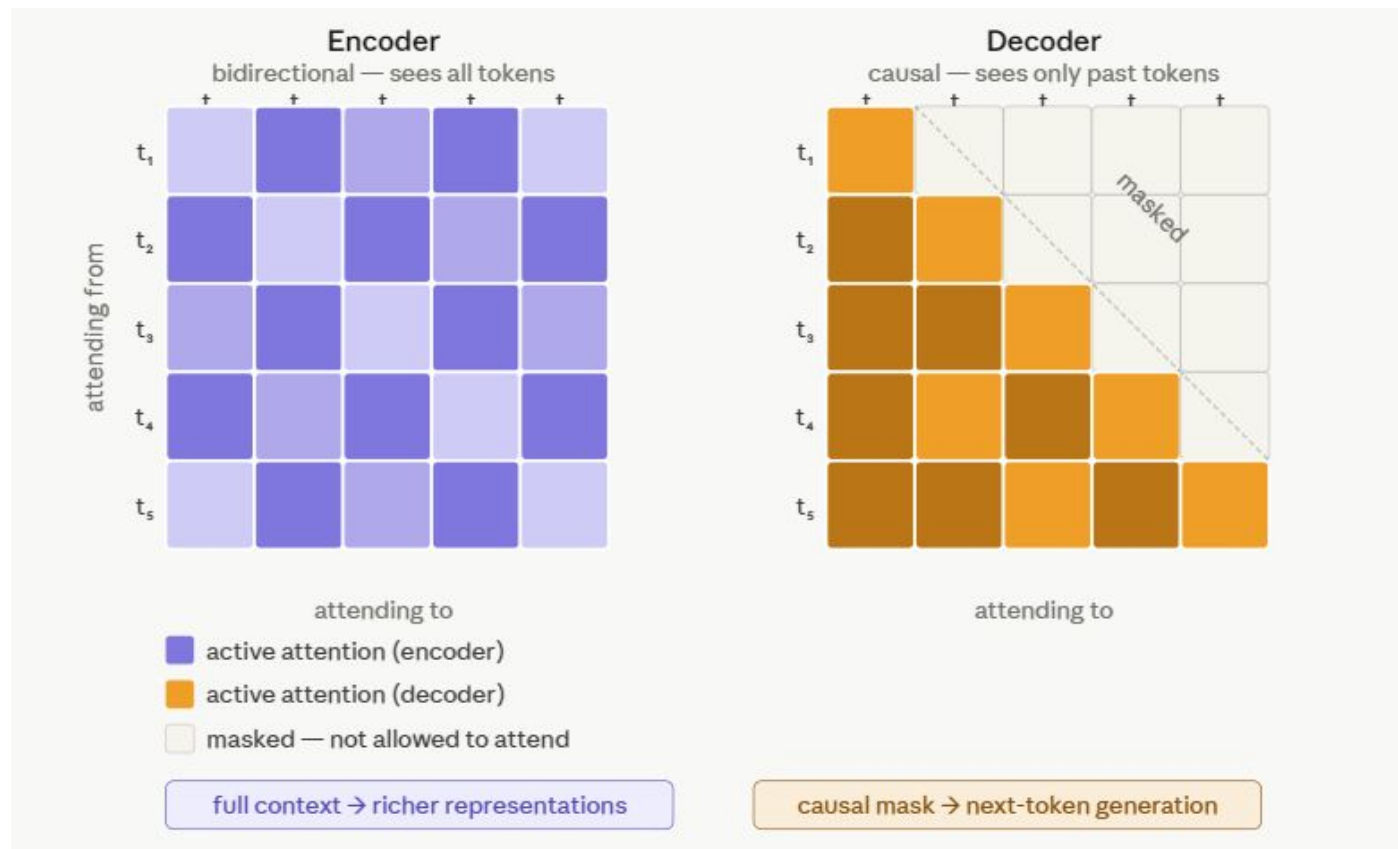
Multimodal transformers... what ARE they?

Transformer based model that operate on multiple covariates

For example GLUMIND

Encoder based transformer that uses cross attention and multiscale attention plus learn without forgetting algorithm

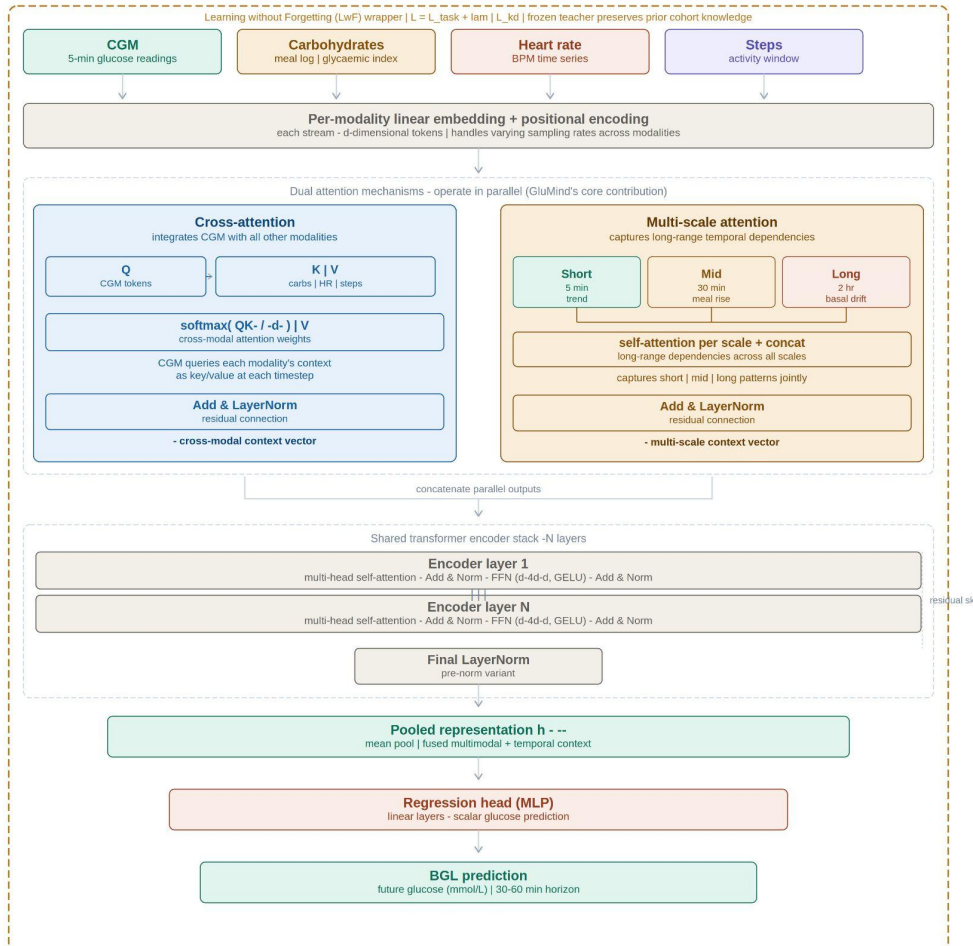
Glumind- Encoder based transformer-Why?



Glumind- Encoder based transformer-General schema

Token embedding+positional encoding

Cross attention and multiscale attention



Let's get back to basics- HOW data is converted

CGM



token1

token2

token3

token4

token5

Option 1-

Patch tokenisation

HR

+positional encoding

steps



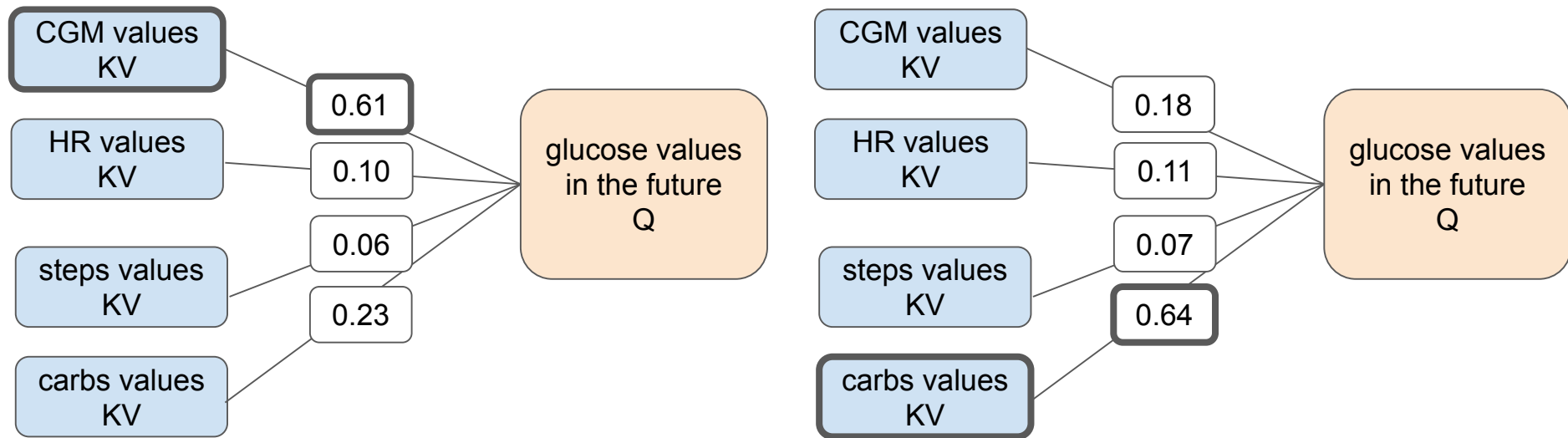
carbs



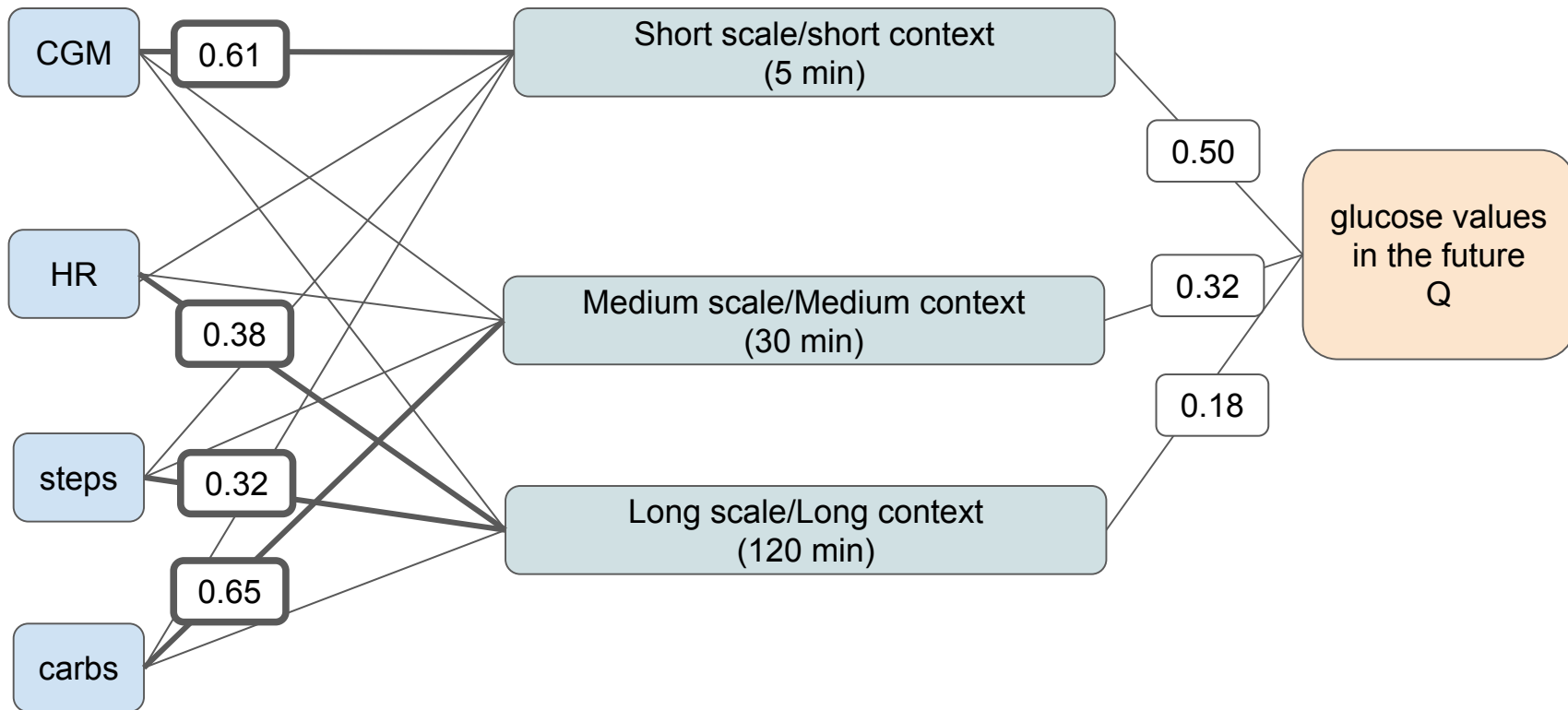
Option 2-

Multiscale convolution

Glumind- Cross-attention



Glumind- Multiscale-attention



Glumind- Learn without forgetting -The Issue

THE PROBLEM: Naive fine-tuning overwrites prior cohort knowledge

Cohort 1

Healthy

Cohort 2

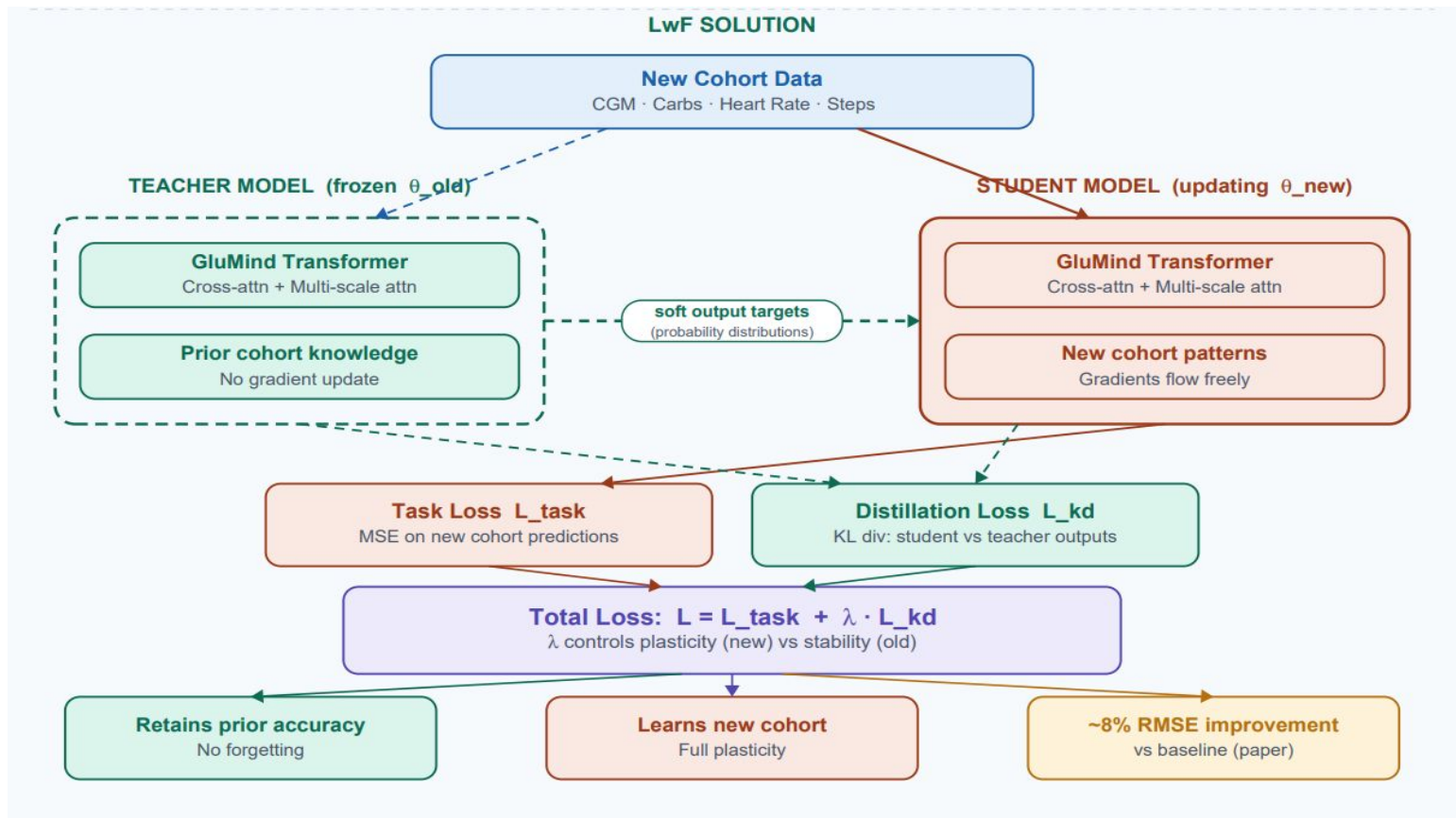
Prediabetes

Cohort 3

Type 2 Diabetes

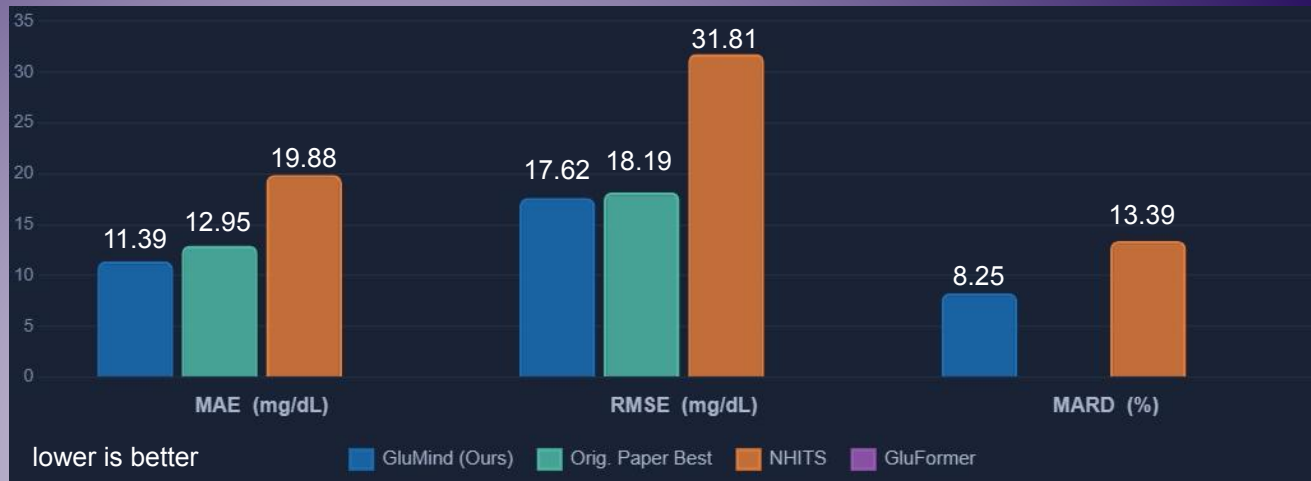
Without LwF: gradients from each new cohort overwrite weights → Catastrophic Forgetting

Glumind- Learn without forgetting -The solution



RESULTS?

For type 2
diabetics and
prediabetes



For type 1
diabetics

